FRANCESCO MORI

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PROFESSIONAL SUMMARY

I am a theoretical physicist at Harvard University, supported by the Center of Mathematical Sciences and Applications and a Shuman Educational Research Grant. My research focuses on the dynamics and the control of nonequilibrium systems, with applications to active matter and machine learning.

RESEARCH EXPERIENCE

Research Associate (independent postdoctoral position) *Sept. 2025 - Present*
Center of Mathematical Sciences and Applications, Harvard University.

Leverhulme-Peierls Fellow (independent postdoctoral position) *Oct. 2022 - Sept. 2025*
Rudolf Peierls Centre for Theoretical Physics, Department of Physics, University of Oxford.

Junior Research Fellow, New College, Oxford. *Oct. 2022 - Sept. 2025*

Part-time Consultant, Scroll Prize, Inc. *Sept. 2024 - June 2025*
Contributing to the Vesuvius Challenge. Image reconstruction of ancient papyri (pre-79 AD).

Ph.D. in Theoretical Physics, Université Paris-Saclay *Oct. 2019 - June 2022*
Laboratory of Theoretical Physics and Statistical Models (LPTMS), Orsay.
Supervisor: Satya Majumdar.
Title: *Extreme value statistics of stochastic processes: from Brownian motion to active particles*.

TEACHING

Abilitazione Scientifica Nazionale *2025*
Accredited to hold Associate Professor positions in Italian universities. (Sections 02/A2 and 02/B2)

Qualification aux fonctions de maître de conférences *2024*
Accredited to hold lecturer positions in French universities. (Section 28 - Theoretical Physics)

Stipendiary Lecturer, New College (Oxford) *2023*
Mathematical Methods, Thermal Physics.

Tutor, Oxford Study Abroad Program *2023*
Biological Physics.

Teaching Assistant, Université Paris-Saclay *2021 - 2022*
Computer Science, Statistical Physics.

AWARDS

European Physical Society - Statistical and Nonlinear Physics Division Early Career Prize *2025*
for outstanding contributions on the statistical properties of run and tumble particles, thermodynamic cost of stochastic resetting and optimal control protocols.

Shuman Educational Research Grant *9-month research grant.* *2025*
Award converted from a **Fulbright Scholarship** *that I was unable to take up due to visa constraints.*

EDUCATION

M. Sc. in Physics of Complex Systems, Université Paris-Saclay *Sept. 2018 - Jul. 2019*
Ranking: 1/42, GPA: 18.6/20

M. Sc. in Physics of Complex Systems, Politecnico di Torino *Oct. 2017 - Jul. 2019*

GPA: 30.00/30, Final mark: 110/110 cum laude.

M. Sc. in Engineering Physics, Politecnico di Milano *Oct. 2017 - Jul. 2019*
Final mark: 110/110 cum laude.

B. Sc. in Applied Mathematics, Politecnico di Torino *Oct. 2014 - Jul. 2017*
GPA: 29.29/30, Final mark: 110/110 cum laude.

INTERNSHIPS AND VISITING POSITIONS

Research Visitor, City University of New York. *Feb. 2025 - Present*

Research Intern, LPTMS, Orsay (with Satya Majumdar). *Mar. 2019 - Jun. 2019*

iMat Project (Project on natural language processing and materials science) *Jun. 2018 - Sept. 2019*
European Materials Modelling Council, Alta Scuola Politecnica.

Visiting Student, SISSA and ICTP (Trieste, Italy). *Sept. 2017 - Feb. 2018*

Visiting Student, Lund University (Sweden). *Aug. 2016 - Feb. 2017*

SCHOLARSHIPS AND TRAVEL GRANTS

Lockey Fund Award (USA) *Travel award to attend scientific conferences in the USA.* *2024*

Lockey Fund Award (Europe) *Travel award to attend scientific conferences in Europe.* *2024*

Astor Travel Scholarship *Travel fund for visits to the USA.* *2024*

Université Paris-Saclay International Master's Scholarship *2018*
One-year master's program at Université Paris-Saclay.

Erasmus Scholarship *2018*
6-month exchange program at Paris-Saclay University.

Alta Scuola Politecnica *2017*
Educational program for the top 1% of master students at Politecnico di Torino and Milano.

Physics of Complex Systems Travel Grant *2017*
6-month exchange program at SISSA and ICTP (Trieste, Italy).

Young Talent Project Travel Grant *2016*
6-month exchange program at Lund University (Sweden).

Young Talent Project *2014*
Educational program for the top 5% of bachelor's students at Politecnico di Torino.

MENTORSHIP

Yaprak Onder (Oxford undergraduate) *2023*
Currently Ph.D. student at Harvard University.

Costantino Di Bello (Université Paris-Saclay master's) *2021*
Currently postdoctoral researcher at SISSA (Trieste).
This internship resulted in the publication Phys. Rev. E **108**, 014112 (2023).

Marco Biroli (École normale supérieure de Paris master's) *2021*
Currently postdoctoral fellow at UChicago.
This internship resulted in the publication J. Phys. A **55**, 244001 (2022).

ACADEMIC SERVICE AND OUTREACH

Organizer, Harvard CMSA Colloquia *2025*

Volunteer, Squishy Science Sunday (outreach activity at the APS March Meeting) *Mar. 2025*

Assessor for master's project *Apr. 2024*
Oxford Interdisciplinary Bioscience DTP

Reviewer *Mar. 2021 - Present*
SciPost, Cambridge University Press, Nat. Commun., PRL, PRE, J. Phys. A: Math. Theor., J. Stat. Mech., Physica A.

Interviewer, University College (Oxford) *Dec. 2022*
Undergraduate Physics admissions

Organizer, Cross-TP discussions *Oct. 2022 - Mar. 2023*
Journal club across all areas of Theoretical Physics in Oxford

Organizer, Fête de la science (outreach activity for high-school students) *Oct. 2021*

PUBLICATIONS (* KEY PAPERS)

26. **(*)** B. Bordelon and **F. Mori**, preprint arXiv:2602.04774 (2026).
25. R. J. Ewart, P. Reichherzer, S. Ren, S. Majeski, **F. Mori**, M. L. Nastac, A. F. A. Bott, M. W. Kunz, A. A. Schekochihin, "Cosmic-ray transport in inhomogeneous media", Mon. Not. R. Astron. Soc. **545**, 2 (2026).
24. **F. Mori** and F. Mignacco. "Analytic theory of dropout regularization", Phys. Rev. E **112**, 045301 (2025).
23. **(*)** F. Mignacco and **F. Mori**, "A statistical physics framework for optimal learning", preprint arXiv:2507.07907 (2025).
22. **(*)** **F. Mori**, S. Sarao Mannelli, and F. Mignacco. "Optimal Protocols for Continual Learning via Statistical Physics and Control Theory," ICLR 2025 and J. Stat. Mech. 084004 (2025).

    - Accepted for poster presentation at the NeurIPS 2024 workshop Mathematics of Modern Machine Learning.

    - Accepted for poster presentation at COSYNE 2025.
21. **(*)** **F. Mori** and L. Mahadevan, "Optimal switching strategies for navigation in stochastic settings", J. R. Soc. Interface 22 (227), 20240677 (2025).
20. **F. Mori**, S. N. Majumdar, and P. Vivo. "Cost of excursions until first crossing of the origin for random walk and Lévy flights: An exact general formula", Phys. Rev. Research **6**, 043053 (2024).
19. K. S. Olsen, D. Gupta, **F. Mori**, S. Krishnamurthy, "Thermodynamic cost of finite-time stochastic resetting", Phys. Rev. Research **6**, 033343 (2024).
18. A. Mummery, **F. Mori**, and S. Balbus, "The dynamics of accretion flows near to the innermost stable circular orbit", Mon. Not. R. Astron. Soc. **529**, 1900 (2024).
17. **(*)** **F. Mori**, S. Bhattacharyya, J. M. Yeomans, and S. P. Thampi, "Viscoelastic confinement induces periodic flow reversals in active nematics", Phys. Rev. E **108**, 064611 (2023).
16. S. N. Majumdar, **F. Mori**, and P. Vivo, "Nonlinear-Cost Random Walk: exact statistics of the distance covered for fixed budget", Phys. Rev. E **108** (6), 064122 (2023).
15. C. Di Bello, A. K. Hartmann, S. N. Majumdar, **F. Mori**, A. Rosso, and G. Schehr, "Current fluctuations in stochastically resetting particle systems", Phys. Rev. E **108**, 014112 (2023). **Highlighted as an Editors' Suggestion.**

14. S. N. Majumdar, **F. Mori**, and P. Vivo, “The cost of diffusion: nonlinearity and giant fluctuations”, **Phys. Rev. Lett. 130**, 237102 (2023).
13. **(*)** B. De Bruyne and **F. Mori**, “Resetting in Stochastic Optimal Control”, Phys. Rev. Research **5**, 013122 (2023).
12. **(*)** **F. Mori**, K. S. Olsen, and S. Krishnamurthy, “Entropy production of resetting processes”, Phys. Rev. Res. **5**, 023103 (2023).
11. **F. Mori**, S. N. Majumdar, and G. Schehr, “Time to reach the maximum for a stationary stochastic process”, Phys. Rev. E **106**, 054110 (2022).
10. M. Biroli, **F. Mori**, and S. N. Majumdar, “Number of distinct sites visited by a resetting random walker”, J. Phys. A: Math. Theor. **55**, 244001 (2022).
9. **F. Mori**, G. Gradenigo, and S. N. Majumdar, “First-order condensation transition in the position distribution of a run-and-tumble particle in one dimension”, J. Stat. Mech. 103208 (2021).
8. **(*)** **F. Mori**, S. N. Majumdar, and G. Schehr, “Distribution of the time of the maximum for stationary processes”, Europhys. Lett. **135**, 30003 (2021). **Highlighted as an Editors’ Choice.**
7. **F. Mori**, P. Le Doussal, S. N. Majumdar, and G. Schehr, “Condensation transition in the late-time position of a run-and-tumble particle”, Phys. Rev. E **103**, 062134 (2021).
6. S. N. Majumdar, **F. Mori**, H. Schawe, and G. Schehr, “Mean perimeter and area of the convex hull of a planar Brownian motion in the presence of resetting”, Phys. Rev. E **103**, 022135 (2021).
5. **F. Mori**, P. Le Doussal, S. N. Majumdar, and G. Schehr, “Universal properties of a run-and-tumble particle in arbitrary dimension”, Phys. Rev. E **102**, 042133 (2020). **Highlighted as an Editors’ Suggestion.**
4. B. Lacroix-A-Chez-Toine, **F. Mori**, “Universal survival probability for a correlated random walk and applications to records”, J. Phys. A: Math. Theor. **53**, 495002 (2020).
3. **(*)** **F. Mori**, P. Le Doussal, S. N. Majumdar, and G. Schehr, “Universal survival probability for a d-dimensional run-and-tumble particle”, **Phys. Rev. Lett. 124**, 090603 (2020).
2. **F. Mori**, S. N. Majumdar, and G. Schehr, “Distribution of the time between maximum and minimum of random walks”, Phys. Rev. E **101**, 052111 (2020).
1. **F. Mori**, S. N. Majumdar, and G. Schehr, “Time between the maximum and the minimum of a stochastic process”, **Phys. Rev. Lett. 123**, 200201 (2019).

INVITED TALKS

Statistical Physics & Machine Learning: moving forward *2025*
Institut d’Études Scientifiques de Cargèse (France).

Paris Biological Physics Community Day *2024*
École normale supérieure (Paris).

Workshop: Stochastic Systems in Active Matter *2024*
Isaac Newton Institute (Cambridge).

Workshop: New Vistas in Stochastic Resetting *2024*
The Higgs Centre for Theoretical Physics (Edinburgh).

Saturday Mornings of Theoretical Physics (outreach activity for Oxford Physics alumni) *2023*
University of Oxford.

Theoretical Physics Colloquium *2022*
University of Oxford.

Large Deviations, Extremes and Anomalous Transport in Non-equilibrium Systems *2022*

The Erwin Schrödinger International Institute for Mathematics and Physics (Austria).

Nordita Scientific Program "Are there universal laws in nonequilibrium physics" *2022*
Nordita Institute, Stockholm (Sweden).

INVITED SEMINARS

MIT Physics of Living Systems Seminar *2026*
Massachusetts Institute of Technology.

LPTMC seminars *2022, 2023, and 2024, 2026*
Laboratoire de Physique Théorique de la Matière Condensée, Paris.

LPTMS Seminar *2026*
Laboratory of Theoretical Physics and Statistical Models, Orsay.

MIT Physics of Living Systems Short Talk *2025*
Massachusetts Institute of Technology

Mathematical Physics Group Seminar *2025*
University of Bologna.

Disordered System Seminar *2022 and 2025*
King's College London.

Condensed Matter Seminar *2025*
University of Massachusetts Amherst.

Soft Matter Seminar *2025*
University of California, Berkeley.

ML Nosh Lunch Seminar *2025*
University of Oxford.

Soft Matter Group Away Day *2024*
University of Oxford.

Soft Matter Seminar *2023*
University of California, Santa Barbara.

Soft Condensed Matter Seminar *2023*
New York University.

IPhT Seminar *2023*
Institut de Physique Théorique, Saclay.

LOMA Seminar *2023*
Laboratoire Ondes et Matière d'Aquitaine, Bordeaux.

Statistical Physics and Complexity Webinar Series *2022*
University of Edinburgh.

LuxStatMech seminar *2022*
University of Luxembourg.

SIFS Young Seminar *2022*
Italian Society of Statistical Physics.

ICTS Statistical Physics Journal Club *2021*
International Centre for Theoretical Sciences, Bangalore.

CONTRIBUTED TALKS

Journée "Physique et Vivant" *2023*
Institut Jacques Monod (Paris).

Nordita Workshop: Fluctuations and First-Passage Problems *2023*
Nordita Institute, Stockholm (Sweden).

4th Course on Multiscale Integration in Biological Systems *2021*
Institut Curie, Paris (France).

Journée Systèmes & Matière Complexes *2021*
Université Paris-Saclay, Paris (France).